[INCH-POUND]
A-A-59500
2 December 1999

COMMERCIAL ITEM DESCRIPTION

BLADES, SHEARING MACHINE, METAL SQUARING

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers replacement blades for metal squaring, shearing machines.

2. CLASSIFICATION. The shearing machine blades shall be of the following types and grades. Grade refers to the tool steel alloy designation per American Society for Testing and Materials (ASTM) E527, "Standard Practice for Numbering Metals and Alloys (UNS)", and ASTM A681, "Standard Specification for Tool Steels Alloy", as shown in table I. Refer to section 7.4 for general guidelines to blade type selection. The type and grade to be furnished shall be as specified (see 7.2(b)).

Type I - 2 cutting edges
Type II - 4 cutting edges

3. SALIENT CHARACTERISTICS

3.1 General requirements. The shear blades shall be manufactured in accordance with the requirements shown in table I. Unless otherwise specified (see 7.2(c)), the blades shall meet the hardness requirements shown in table II when tested in accordance with ASTM E18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials". Unless otherwise specified (see 7.2(d)), the blade tolerances and dimensions shall be in accordance with the original equipment manufacturer's (OEM) requirements for the make and model of shearing machine specified (see 7.2(e)).

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3455

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TABLE I. Blade grades and alloy content requirements.

Grade	ASTM E527 Unified number (ASTM A681 type designation)	Percent (%) of alloy element content per table 1 of ASTM A681								
		C	Mn	Si	Cr	V	W	S	P	Mo
A	T30402 (D2)	1.40 - 1.60	0.10 - 0.60	0.10 - 0.60	11.0 - 3.0	0.50 - 1.10		0.03 max.	0.03 max.	0.70 - 1.20
B	T30403 (D3)	2.00 - 2.35	0.10 - 0.60	0.10 - 0.60	11.0 - 13.0	1.00 max.	1.00 max.	0.03 max.	0.03 max.	
C	T30102 (A2)	0.95 - 1.05	0.40 - 1.00	0.10 - 0.50	4.75 - 5.50	0.15 - 0.50		0.03 max.	0.03 max.	0.90 - 1.40
D	T41904 (S4)	0.50 - 0.65	0.60 - 0.95	1.75 - 2.25	0.10 - 0.50	0.15 - 0.35		0.03 max.	0.03 max.	
E	T41905 (S5)	0.50 - 0.65	0.60 - 1.00	1.75 - 2.25	0.10 - 0.50	0.15 - 0.35		0.03 max.	0.03 max.	0.20 - 1.35
F	T41906 (S6)	0.40 - 0.50	1.20 - 1.50	2.00 - 2.50	1.20 - 1.50	0.20 - 0.40		0.03 max.	0.03 max.	0.30 - 0.50
G	T41907 (S7)	0.45 - 0.55	0.20 - 0.90	0.20 - 0.90	3.0 - 3.50	0.35 max.		0.03 max.	0.03 max.	1.30 - 1.80

Alloy elements: C-Carbon, Mn-Manganese, Si-Silicon, Cr-Chromium, V-Vanadium, W-Tungsten, S-Sulphur, P-Phosphorus, and Mo-Molybdenum

TABLE II. Blade hardness requirements, Rockwell C-scale.

Grade	Thickness of metal being sheared, inches									
	0.063	0.188	0.250	0.375	0.500	0.625	0.750	1.000	1.250	1.500
A & B	59-61	59-61	57-59	57-59	–	–	–	–	–	–
C	57-62	57-62	57-62	57-62	57-59	57-59	57-59	–	–	–
D & E	–	–	53-56	53-56	53-56	51-55	51-55	50-54	50-54	50-54
F & G	–	–	–	53-56	53-56	51-55	51-55	48-54	48-54	48-54

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the product shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(f)).

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7. NOTES

7.1 Source of documents.

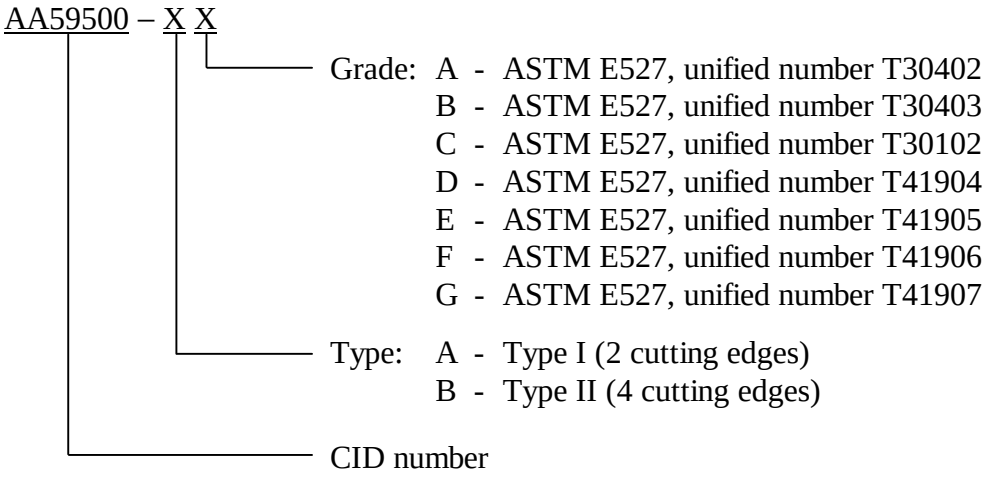
7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. The type and grade of blade required (see 2)
c. Rockwell hardness, if different (see 3.1)
d. Blade tolerances and dimensions, if different (see 3.1)
e. Shearing machine make and model, for use on (see 3.1)
f. Packaging requirements (see 6.1)

7.3 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



7.4 Blade type and application. The correct blade type and heat treatment for any plate shearing application is primarily dependent upon the chemical and physical properties of the work material being sheared. Table III contains general guidelines for blade type selection. The user should review shearing applications with the original equipment manufacturer of the shearing machine for selecting the correct blade type and hardness.

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TABLE III. Recommended blade grade(s) for metal shearing applications.

Application	Thickness of metal being sheared (in)									
	0.063	0.188	0.250	0.375	0.500	0.625	0.750	1.000	1.250	1.500
General machine shop/Mild steel (1010-1023, A283, A285, A36)	A	A	A B	C	C D	E	E F	E F	F G	F G
High strength, low alloy steel (A242, A-441, A-572, A-573, A-588, A-633, A-808)	N/A	A	A B	C	C D	E	E F	E F	F G	F G
Quenched and tempered low carbon alloy steel (A-514, A-517, A-678)	N/A	A	C	C D	E	E	E F	F G	F G	F G
Abrasion resistant steel (360 Bhn, type AR, RQ)	N/A	C	C D	E	E	F	F G	F G	N/A	N/A
Stainless (414, 431, 440, 301)	A	A	C D	C D	E	F	F G	F G	G	G
Stainless (416, 418, 446, 302, 304, 316, 317, 308, 309)	A	A	C D	C D	E	F	F G	F G	G	G
Aluminum, temper 0 (1100, 2024, 5052, 6061)	A	A	A B	A B	B	B	C D	E	E	E
Aluminum (7075T6, 2024T3, 2024T4)	A	A	A B	B	C	C	C D	E	E	E
Copper	A	A	A B	A B	C	C	C	E	E	E
Titanium (Alloy types 5 and 6)	A	B	C D	C D	E	E F	E F	F G	G	G
Inconel 718	A	C	C D	C D	E	E F	E F	F G	N/A	N/A
Hastelloy	A	C	C D	C D	E	E F	E F	F G	N/A	N/A

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL

Reviewer:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3455-0160)

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